Kpodjro KPATOUKPA

Data Scientist

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Data Scientist specializing in generative AI and deep learning, with two years of experience working on RAG systems and scoring models in production. I am skilled at taking on complex projects in machine learning and generative AI.

PROFESSIONAL EXPERIENCE

Generative AI Data Scientist 10/2025 – 09/2026
Innovation & Digital Directorate - URSSAF, Montreuil, France

- Performed prompt engineering and developed an advanced RAG chatbot integrating heterogeneous document.
- Built OCR and image enhancement pipelines for scanned documents using LightOn-OCR and PaddleOCR.
- Designed and implemented an automated evaluation protocol to objectively benchmark RAG system performance.
- Technology watch and experimentation on advanced RAG architectures (GraphRAG, LightRAG, Neo4j).

Machine Learning Engineer 03/2024 – 08/2024
ATTIJARIWAFA BANK, Casablanca

- Extracted and preprocessed data from PDF and text files, including embedded image processing.
- Developed an augmented RAG system to handle B2B customer queries at scale.
- Fine-tuned a Mistral model and benchmarked its performance against the RAG-based approach.
- Delivery of a chatbot interface and monitoring dashboard (4 KPIs) adopted by the business team.

Junior Data Scientist 08/2023 – 09/2023
MLView Consulting, Tangier

- Performed data analysis and extracted marketing insights in collaboration with business stakeholders.
- Trained a deep learning model for customer targeting in future marketing campaigns.
- Fine-tuned the Llama2 LLM to generate personalized marketing copy.

TECHNICAL SKILLS

- **GenerativeAI :** LLM, LangChain, LlamaIndex, RAG, GraphRAG, Fine-tuning, DeepEval, LLM-as-judge.
- **Machine Learning :** Scikit-learn, PyTorch, Torchvision, Time Series, Clustering, Réduction de dimension.
- **NLP & Vision :** LLM, EasyOCR, PaddleOCR, MistralOCR, ADE, LightOn-OCR, perspective transformation.
- **Data, web & Cloud :** Python, SQL, NoSQL, BigQuery, GCP, R, Selenium, Scrapy, BeautifulSoup, Neo4J, Cypher.
- **Dev & Dataviz :** Docker, Git, FastAPI, Streamlit, RShiny, PowerBI, Tableau, Seaborn, Java, C++, NetworkX.

CROSS-DISCIPLINARY SKILLS

Soft Skills : Teamwork & collaboration, Adaptability, Analytical thinking, Intellectual curiosity.
Languages : French (Native), English (Professional - C1).

EDUCATION

M.Sc. Machine Learning for Data Science 09/2024 – 08/2026
Université Paris Cité, Paris, France

- Reinforcement Learning, Graph Learning, Recommendation Systems, Clustering & Dimensionality Reduction, NLP, Data Visualization, Time Series, Mixture Models, Vertex AI (GCP).

M.Eng. Intelligent Systems Engineering 10/2019 – 07/2024
Faculty of Sciences and Technology, Tangier, Morocco

- Machine Learning, Deep Learning, AI Methodology, Computer Vision, Data Mining, Inferential Statistics, BI.

RESEARCH PROJECTS

- **Categorical Generative Neural Networks (10/2025 – 12/2025) :** Built a synthetic data generation model using a Categorical GAN (CatGAN, scikit-learn) to augment training datasets – achieving 97% reliability.
- **Recommendation System (10/2024 – 05/2025) :** Implemented a hybrid recommendation engine combining generative AI and graph-based methods (LightGNN, Matrix Factorization, Ensemble Models, XGBoost).
- **Demand Forecasting :** Developed a supply chain forecasting model for supermarkets, reaching 91% accuracy.